

Homework Set 5

Problem 1: We would like to apply off-policy Monte Carlo evaluation to a policy in the infinite-horizon, discounted setting. Consider an MDP with three states $\mathcal{S} = \{1, 2, 3\}$, initial state 1, two actions $\mathcal{A} = \{g, h\}$, transition function $P(s'|s, a)$ defined as

$$\begin{aligned} P(1|1, g) &= 0.1, & P(2|1, g) &= 0.8, & P(3|1, g) &= 0.1, \\ P(1|1, h) &= 0.8, & P(2|1, h) &= 0.1, & P(3|1, h) &= 0.1, \\ P(1|2, g) &= 0.1, & P(2|2, g) &= 0.1, & P(3|2, g) &= 0.8, \\ P(1|2, h) &= 0.1, & P(2|2, h) &= 0.8, & P(3|2, h) &= 0.1, \\ P(1|3, g) &= 0.8, & P(2|3, g) &= 0.1, & P(3|3, g) &= 0.1, \\ P(1|3, h) &= 0.1, & P(2|3, h) &= 0.1, & P(3|3, h) &= 0.8, \end{aligned}$$

reward function $R(s, a)$ outputting 1 for $(3, h)$ and 0 otherwise, and discount factor 0.95.

1. Consider the target policy π^t

$$\begin{aligned} \pi^t(g|1) &= 0.9, & \pi^t(h|1) &= 0.1, \\ \pi^t(g|2) &= 0.9, & \pi^t(h|2) &= 0.1, \\ \pi^t(g|3) &= 0.1, & \pi^t(h|3) &= 0.9, \end{aligned}$$

and the behavior policy π^b

$$\begin{aligned} \pi^b(g|1) &= 0.85, & \pi^b(h|1) &= 0.15, \\ \pi^b(g|2) &= 0.88, & \pi^b(h|2) &= 0.12, \\ \pi^b(g|3) &= 0.1, & \pi^b(h|3) &= 0.9. \end{aligned}$$

Evaluate the infinite-horizon performance of these two policies, i.e., compute the value functions V^{π^t} and V^{π^b} , assuming access to the model. Report the values for both policies.

2. In order to limit the sample trajectories to finite ones in the Monte Carlo algorithm, compute an effective horizon such that the errors in returns are less than $\epsilon = 0.1$.
3. Use off-policy Monte Carlo evaluation for the target policy to estimate the value function in the initial state $s = 1$. In particular, generate 50 trajectories of length equal to the effective horizon (computed in the previous part), starting from $s = 1$ following the behavior policy π^b in the underlying model. Then, use importance sampling to calculate the estimated value function $\hat{V}^{\pi^t}(1)$ in the initial state.
4. Compute the error between the estimated value function $\hat{V}^{\pi^t}$ from the sample finite trajectories (computed in Part 3) and the true infinite-horizon value function V^{π^t} (computed in Part 1) at state $s = 1$.

[**Note:** For this problem, attach all your code in a programming language of your choice to the end of your submission. You may get help from or use the existing implementations of the Monte Carlo algorithms.]

Problem 2: Consider linear function approximation for representing the state-action value function in the infinite-horizon, discounted setting. In this case, one may define a set of k features $\phi_l(s, a)$ for $l \in \{1, 2, \dots, k\}$ and approximate the state-action value function as a linear combination of these features

$$Q_\theta(s, a) = \theta^\top \phi(s, a) = \sum_{l=1}^k \theta_l \phi_l(s, a)$$

weighted by parameters θ_l . Here θ and $\phi(s, a)$ are column vectors concatenating θ_l and $\phi_l(s, a)$, respectively, for all $l \in \{1, 2, \dots, k\}$.

Example: Suppose in an MDP, the states and actions take real values, that is $\mathcal{S} \subseteq \mathbb{R}$ and $\mathcal{A} \subseteq \mathbb{R}$. One may use a linear function class to form a class of second-order polynomials in s and a by using features $\phi_1 = 1$, $\phi_2 = s$, $\phi_3 = a$, $\phi_4 = sa$, $\phi_5 = s^2$, and $\phi_6 = a^2$. In this case, $\theta = [\theta_1 \ \theta_2 \ \theta_3 \ \theta_4 \ \theta_5 \ \theta_6]^\top$ represents the parameters that need to be learned and $\phi(s, a) = [\phi_1 \ \phi_2 \ \phi_3 \ \phi_4 \ \phi_5 \ \phi_6]^\top$ represents all the features for any (s, a) . With these features, the state-action value function will be of the form $Q_\theta(s, a) = \theta_1 + \theta_2 s + \theta_3 a + \theta_4 sa + \theta_5 s^2 + \theta_6 a^2$. In general, there is a great deal of flexibility in the design of the features that allows one to create essentially any type of function classes. For instance, a feature could be $\phi(s, a) = \mathbb{1}[\alpha < s + a < \beta]$ or $\phi(s, a) = \exp(s)$. In practice, knowing good features is itself challenging, and therefore models such as neural networks can help to learn both good features $\phi(s, a)$ and parameters θ .

1. For an MDP with a finite state space $\mathcal{S}$ and a finite action space $\mathcal{A}$, one can use a tabular representation of the state-action value function $Q(s, a)$ for all states and actions. Show that the tabular representation is a special class of linear functions by creating a class of linear functions

$$\mathcal{Q} = \{Q_\theta : Q_\theta(s, a) = \theta^\top \phi(s, a) \text{ for all } s \in \mathcal{S} \text{ and } a \in \mathcal{A}, \theta \in \mathbb{R}^k\},$$

parameterized by θ that is equivalent to the tabular representation. Determine how the number of features k and the vector-valued function $\phi(s, a) : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^k$ that maps each state-action pair to a k -dimensional feature vector should be defined.

2. Consider an MDP with a finite state space $\mathcal{S}$ and a finite action space $\mathcal{A}$. Suppose we have identified that the state space and the action space can be partitioned based on the similarity between the states and actions:

$$\begin{aligned} \mathcal{S} &= \mathcal{S}_1 \cup \mathcal{S}_2 \cup \dots \cup \mathcal{S}_n \text{ such that } \mathcal{S}_{j_1} \cap \mathcal{S}_{j_2} = \emptyset \text{ for all } j_1 \neq j_2, \\ \mathcal{A} &= \mathcal{A}_1 \cup \mathcal{A}_2 \cup \dots \cup \mathcal{A}_m \text{ such that } \mathcal{A}_{j_1} \cap \mathcal{A}_{j_2} = \emptyset \text{ for all } j_1 \neq j_2, \end{aligned}$$

i.e., the subsets $\mathcal{S}_i$ and $\mathcal{A}_j$ group the similar states and actions (in terms of value functions), respectively. Create a class of linear functions

$$\mathcal{Q} = \{Q_\theta : Q_\theta(s, a) = \theta^\top \phi(s, a) \text{ for all } s \in \mathcal{S} \text{ and } a \in \mathcal{A}, \theta \in \mathbb{R}^k\},$$

parameterized by θ that uses this partitioning to efficiently represent the state-action value function $Q(s, a)$ for all states and actions. Determine how the number of features k and the vector-valued function $\phi(s, a) : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^k$ that maps each state-action pair to a k -dimensional feature vector should be defined.

3. Suppose we want to use supervised learning to approximate the state-action value function $Q^\pi(s, a)$ of an MDP with a large state space $\mathcal{S}$ and large action space $\mathcal{A}$ under policy π . We have collected a data set $\{x^i = (s^i, a^i), y^i\}_{i=1}^N$ of size N , where x^i represents the i^{th} sampled state-action pair and y^i represents a sampled return for x^i . The goal is to learn the best fit to the data by empirical risk minimization with square loss, i.e., solving

$$\hat{Q}^\pi(s, a) = \arg \min_{Q \in \mathcal{Q}} \sum_{i=1}^N (Q(s^i, a^i) - y^i)^2,$$

where

$$\mathcal{Q} = \{Q_\theta : Q_\theta(s, a) = \theta^\top \phi(s, a) \text{ for all } s \in \mathcal{S} \text{ and } a \in \mathcal{A}, \theta \in \mathbb{R}^k\}$$

represents a class of linear functions, parameterized by θ . Assume that $\phi(s, a) : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^k$ is a known vector-valued function that maps each state-action pair to a k -dimensional vector. Also, assume that

$$\sum_{i=1}^N \phi(s^i, a^i) \phi^\top(s^i, a^i)$$

forms an invertible matrix. Derive a closed-form solution for $\hat{Q}^\pi(s, a)$.

[Hint: You can use the fact that this linear parameterization makes the objective function of the minimization problem a convex function in θ such that the solution is a stationary point of the objective function, i.e., a point where the derivative is zero.]

Problem 3: Consider the roll-out procedure for sampling a trajectory and its return from state s and action a in an infinite-horizon, discounted setting. The trajectory is sampled according to the transition function P , the policy π , and the reward function R (assumed to be deterministic). The length of the trajectory (number of time steps) T is sampled from a geometric distribution with parameter $1 - \gamma$, where γ is the discount factor. Prove the following statement discussed in class:

$$\mathbb{E}_{\substack{S_{t+1} \sim P(\cdot | S_t, A_t) \\ A_t \sim \pi(\cdot | S_t) \\ T \sim \text{Geo}(1-\gamma)}} \left[\sum_{t=0}^T \gamma^t R(S_t, A_t) \mid S_0 = s, A_0 = a \right] = Q^\pi(s, a),$$

i.e., the sample return generated through the roll-out procedure creates an unbiased estimator of the state-action value function $Q^\pi(s, a)$.

[Note: Recall that the geometric distribution considered here is defined over $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ such that if $X \sim \text{Geo}(p)$, then $\mathbb{P}[X = k] = (1 - p)^k p$.]

Problem 4: We would like to apply the conservative policy iteration algorithm. We have started with an initial policy π_0 that selects actions over the action space $\mathcal{A} = \{-1, 0, 1\}$ uniformly at random for each state in the state space $\mathcal{S} = \{-3, -2, -1, 0, 1, 2, 3\}$. The class of state-action value function Q we are considering is

$$\mathcal{Q} = \{Q_\theta : Q_\theta(s, a) = \theta_1 s^2 + \theta_2 a^2 + \theta_3 sa + \theta_4, \theta \in \mathbb{R}^4\}.$$

1. Write the objective function of the empirical risk minimization with square loss if we want to fit a Q function to the following data set obtained by the roll-in and roll-out process from policy π_0 :

$$\begin{aligned} s^1 &= 1, a^1 = 0, y^1 = 1 \\ s^2 &= -2, a^2 = 1, y^2 = 0 \\ s^3 &= 1, a^3 = -1, y^3 = 3 \end{aligned}$$

The only variables in the objective function should be $\theta_1, \theta_2, \theta_3$, and θ_4 . There is no need to solve the minimization problem.

2. After collecting a larger data set from the current policy π_0 and solving the empirical risk minimization problem over θ , we have arrived at

$$\hat{Q}^{\pi_0}(s, a) = 2s^2 + a^2 - sa + 0.5.$$

Compute the updated policy π_1 for state $s = -1$, $s = 0$, and state $s = 2$ if the parameter α is set to 0.25.

Problem 5: We aim to find an optimal policy using Q-learning with function approximation, i.e., restricting the state-action value function Q to

$$\mathcal{Q} = \{Q_\theta : \theta \in \mathbb{R}^d\}.$$

In the course of the algorithm applied over an MDP with state space $\mathcal{S} = \{b, c, d, e\}$, action space $\mathcal{A} = \{x, y\}$, and discount factor $\gamma = 0.9$, we have obtained

$$\begin{aligned} Q(b, x) &= -1.5, \quad Q(b, y) = -2.5, \\ Q(c, x) &= -0.5, \quad Q(c, y) = -1.0, \\ Q(d, x) &= 0.0, \quad Q(d, y) = -0.25, \\ Q(e, x) &= -0.5, \quad Q(e, y) = 0.75. \end{aligned}$$

1. Form an ϵ -greedy policy according to this Q function, setting $\epsilon = 0.1$.
2. Using this ϵ -greedy policy, suppose we have collected the following sample sub-trajectory

$$(s_t = c, a_t = x, r_t = 1, s_{t+1} = e, a_{t+1} = x, r_{t+1} = -2, s_{t+2} = b, a_{t+2} = y, r_{t+2} = -0.5, s_{t+3} = b).$$

Create a data set of three data points that can be used to update the parameter θ of the Q function using supervised learning.

3. Let $\mathcal{Q}$ represent a class of linear functions

$$Q_{\theta}(s, a) = \theta^{\top} \phi(s, a) = \sum_{l=1}^2 \theta_l \phi_l(s, a)$$

based on feature $\phi = [\phi_1(s, a) \quad \phi_2(s, a)]^{\top}$ defined as

$$\begin{aligned}\phi_1(s, a) &= -2\mathbb{1}[s = b] - \mathbb{1}[s = c] - 0.5\mathbb{1}[s = d], \\ \phi_2(s, a) &= \mathbb{1}[a = x] - \mathbb{1}[a = y],\end{aligned}$$

and with parameter $\theta = [\theta_1 \quad \theta_2]^{\top} \in \mathbb{R}^2$. Write the objective function of the empirical risk minimization with square loss when fitting a new Q function to the data set in Part 2. The only variables in the objective function should be θ_1 and θ_2 .

4. Solve the optimization problem in Part 3. Report the values the solution yields for parameter θ and the objective function.